Ethanol Blending in Gasoline - Chile

August 2025

U.S. GRAINS & BIOPRODUCTS COUNCIL

20 F St. NW, Suite 900 \ Washington, DC 20001
202-789-0789 \ www.grains.org

FARO90

www.faro90.com
+52 55 6122 2255

Ethanol Blending in Latin America

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Latin America to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.

Sixteen countries with potential and additional use of ethanol were studied: 1) gasoline market profiles; 2) Optimization of gasoline blends with ethanol and 3) Environmental impact of gasolines blended with ethanol.

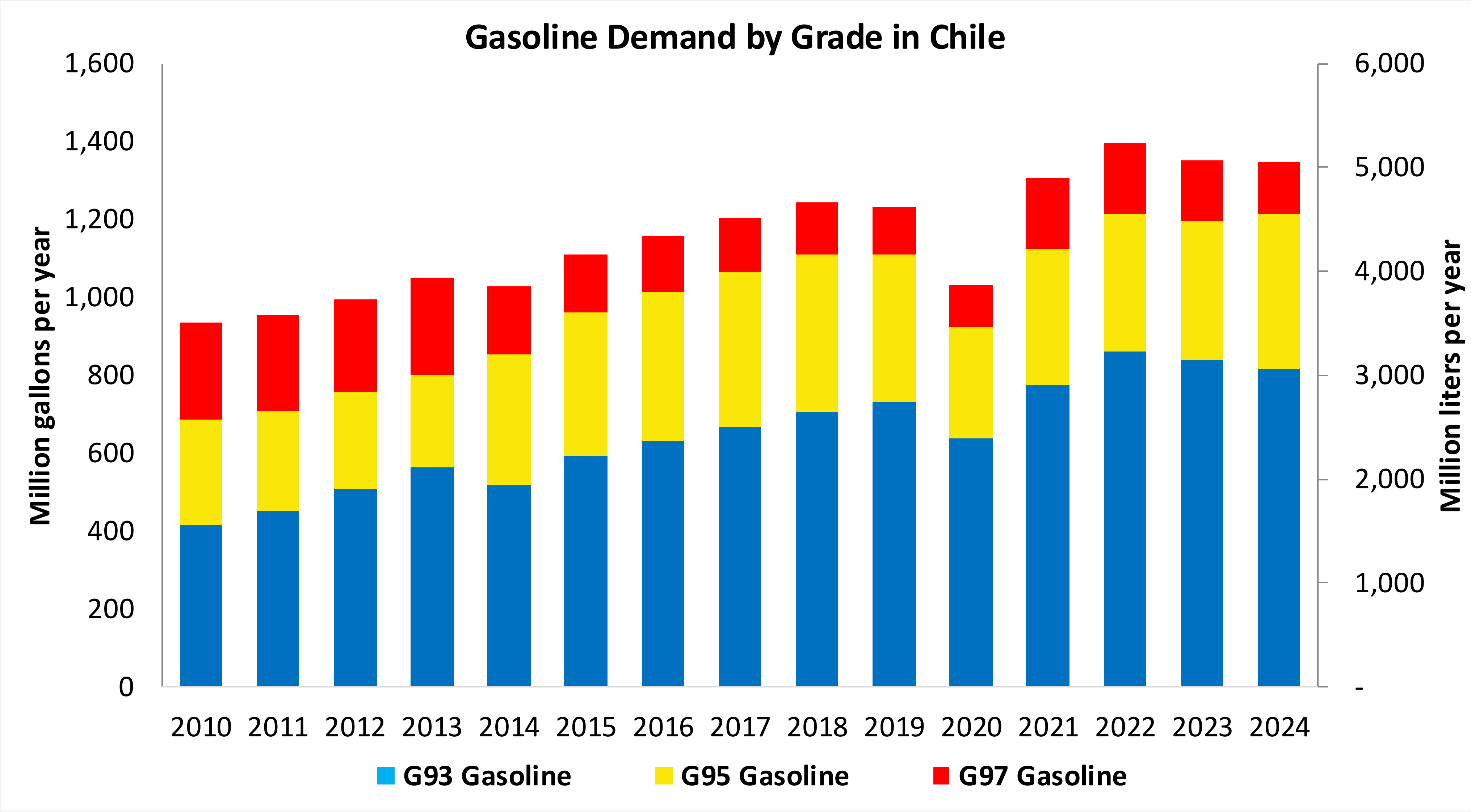
Ethanol Blending in Ethanol - Chile

Chile's gasoline consumption is currently over 1,300 million gallons per year (5,000 million liters) in three grades: RON 93 (AKI 87), RON 95 (AKI 88) and RON 97 (AKI 92). There is a more stringent quality specification for Santiago Metropolitan Area. In 2024, the market share of RON 93 gasoline was 28%. Government company ENAP produces 80% of the total demand (CNE). Gasoline imports made in the country come mainly from the United States. Refineries only produce RON 93 and RON 97, gasoline RON 95 is blended from these grades in gas stations.

Ethanol can be blended up to 5% v/v (Decreto Supremo 56) but it is not used because of regulatory barriers. Chile does not produce or imports ethanol. The use of ethanol is being analyzed to achieve short term goals in the national economy decarbonization program.

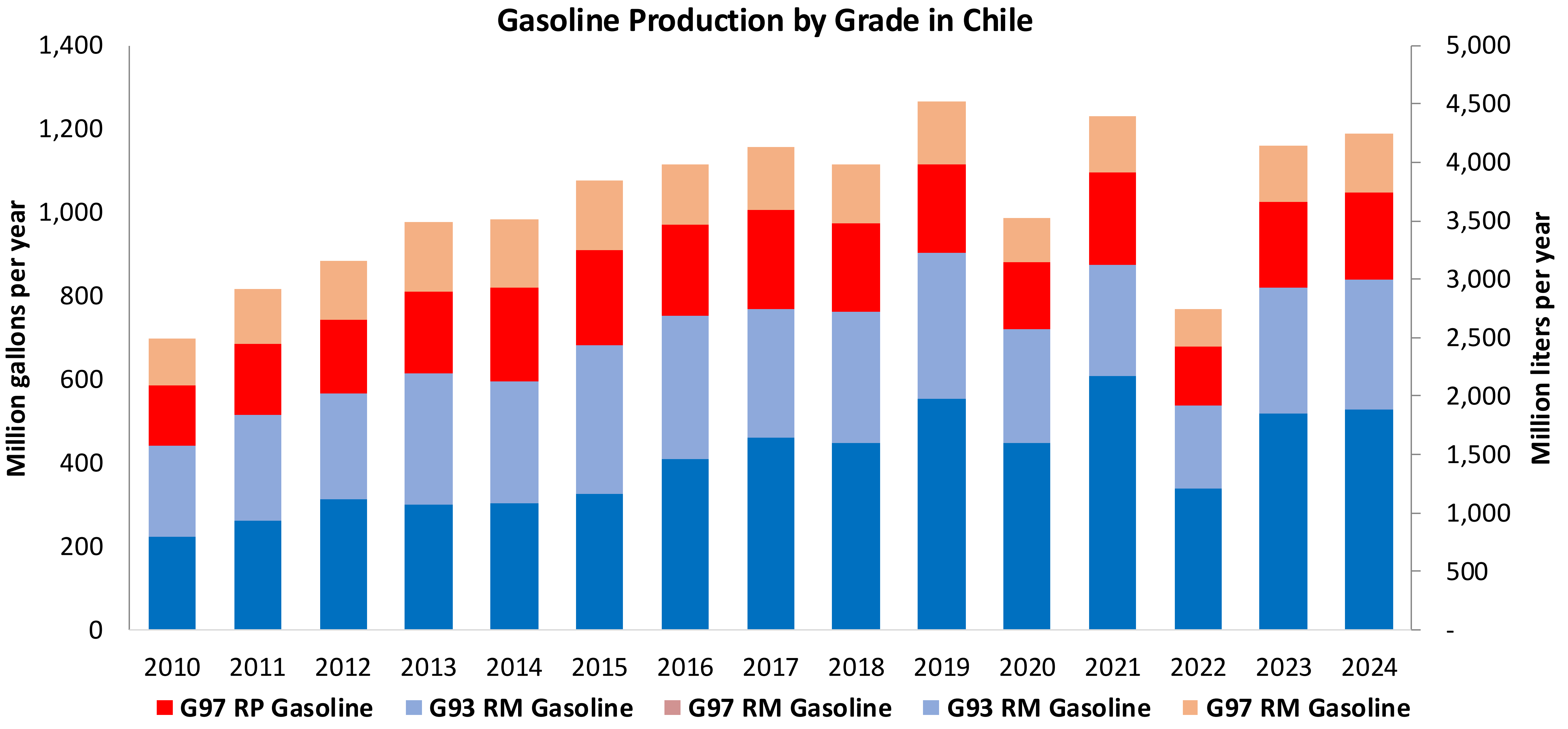
Source: CNE, SEC, 2025

Gasoline Demand in Chile



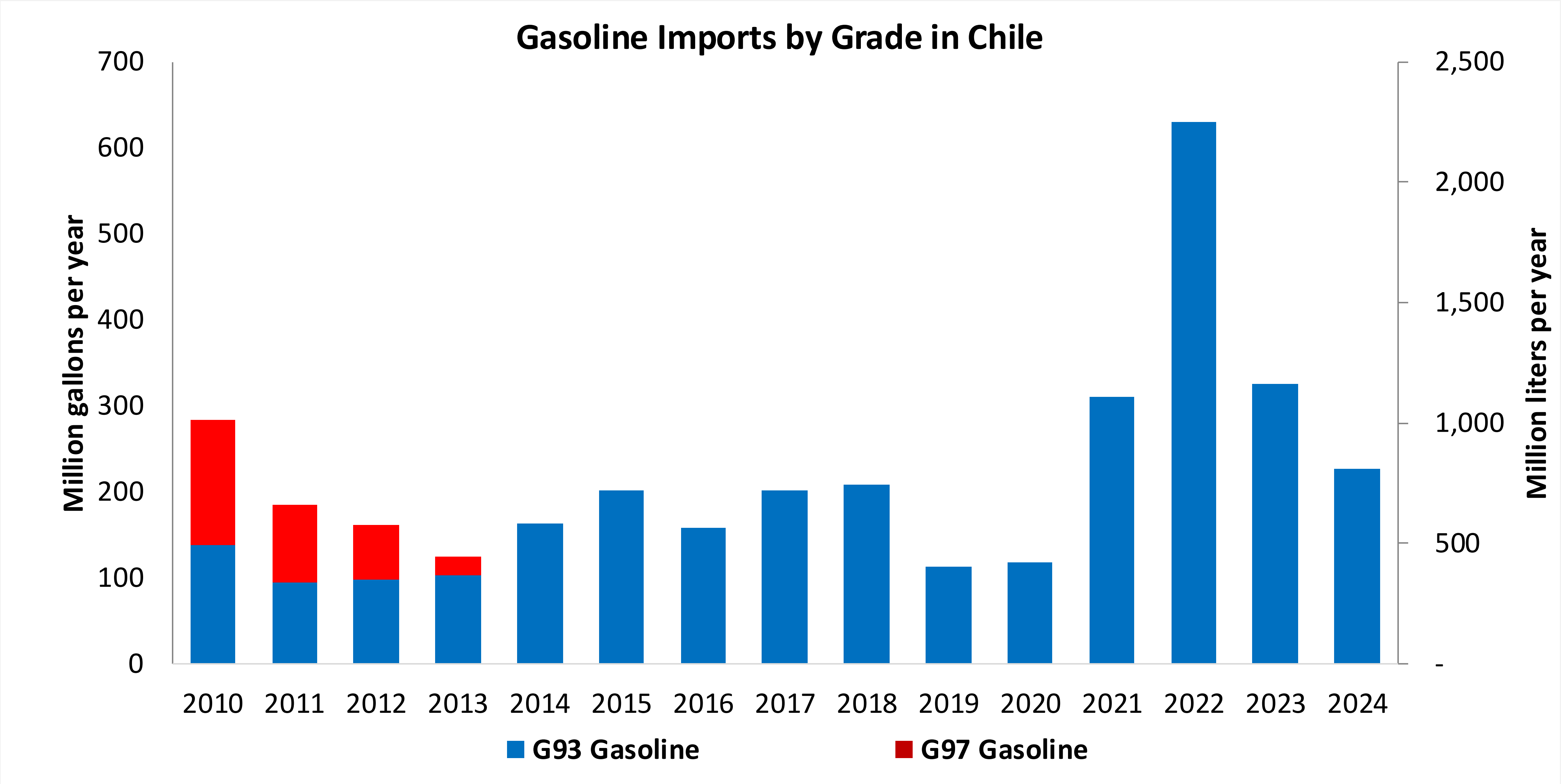
Fuente: CNE, 2025

Gasoline Production in Chile



Fuente: CNE, 2025

Gasoline Imports in Chile



Fuente: CNE, 2025

6

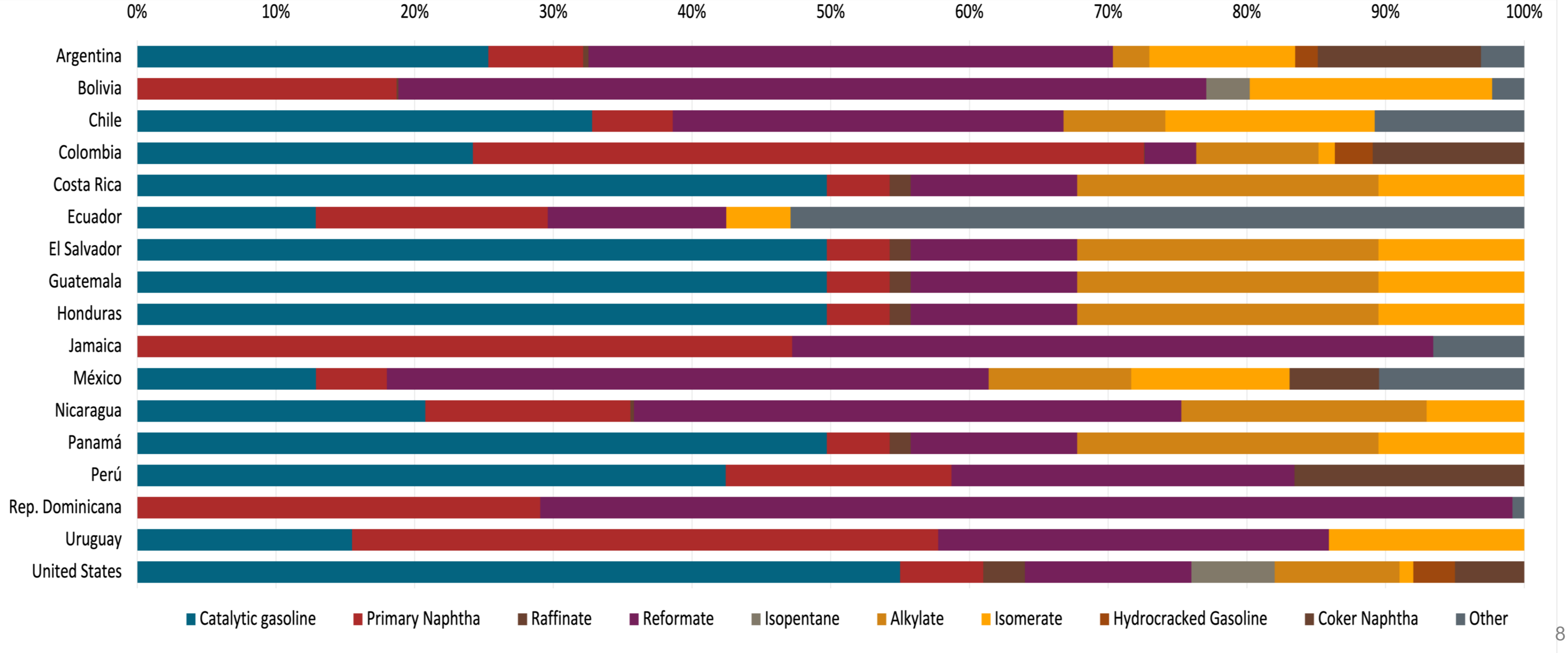
Gasoline Quality in Chile

Name	PPDA DS31	PPDA DS31	DS 60	DS 60	EN 228:2012 + A1:2017 (Euro 6 enabling)			
Implementation Date	2017	2017	2012	2012	2017			
Applicability	Metropolitan Region	Metropolitan Region	Rest of the country	Rest of the country	All countries			
Selected Grade	G93	G97	G93	G97	RON 95 E5	RON 95 E10	RON 98 E5	RON 98 E10
Benzene Content	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v	< 1 %v/v
Aromatics	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 38 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v	< 35 %v/v
Olefins	12 %v/v	12 %v/v	20 %v/v	20 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v	< 18 %v/v
Lead Content	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 0,013 g/l	< 5 mg/l	< 5 mg/l	< 5 mg/l	< 5 mg/l
Manganese	Report	Report	Report	Report	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l	< 2,0 mg/l
RON	93	97	93	97	> 95	> 95	> 98	> 98
MON	Report	Report	Report	Report	> 85	> 88	> 85	> 88
AKI								
Sulfur Content	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 15 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg	< 10 mg/kg
Oxygen Content	< 2 %m/m	< 2 %m/m	< 2 %m/m	< 2 %m/m	<2,7 % m/m	<3,7 % m/m	<2,7 % m/m	<3,7 % m/m
Ethanol (EtOH)	<5 %v/v	<5 %v/v	<5 %v/v	<5 %v/v	<5 %v/v	<10 %v/v	<5 %v/v	<10 %v/v
RVP 37.8°C (Summer)	<> 55 kPa	<> 55 kPa	<> 69 kPa, <> 83 kPa (Magallanes y región Antártica)	<> 69 kPa, <> 83 kPa (Magallanes y región Antártica)	<> 60 - 70 kPa *Depends on the country, RVP is regulated in the EU Fuel Quality Directive			
RVP 37.8 °C(Winter)	<> 69 kPa	<> 69 kPa	<> 45 - 80 kPa	<> 45 - 80 kPa				
RVP 37.8°C (Transition)								
MTBE					-	-	-	-
Ethers 5 or more C Atoms	-	-	-	-	Based on oxygen content	<22 %v/v	Based on oxygen content	<22 %v/v

Source: ENAP, 2022

Gasoline Component Blending in Latin America

Gasoline is a blend of a base gasoline and other components. This blending is usually done at blending terminals as only 30% of the world's finished gasoline is distributed directly from refineries. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Latin America are:



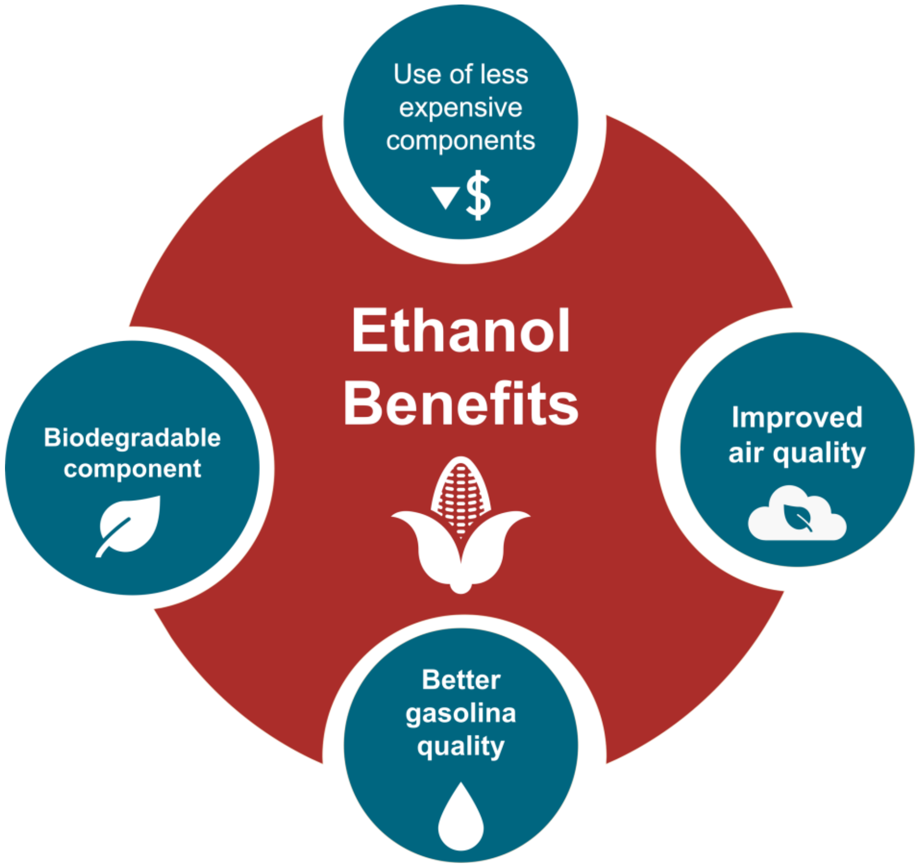
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

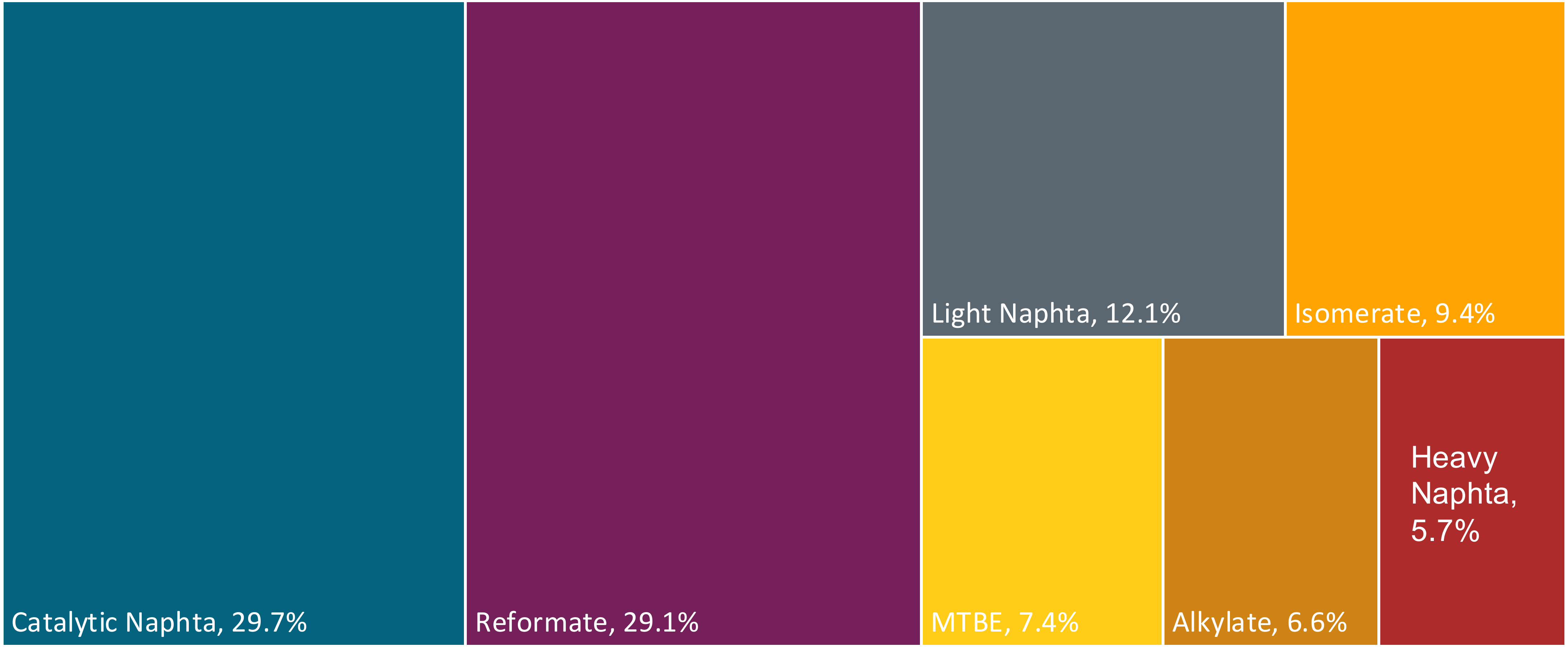
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices January 2022 – February 2023 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

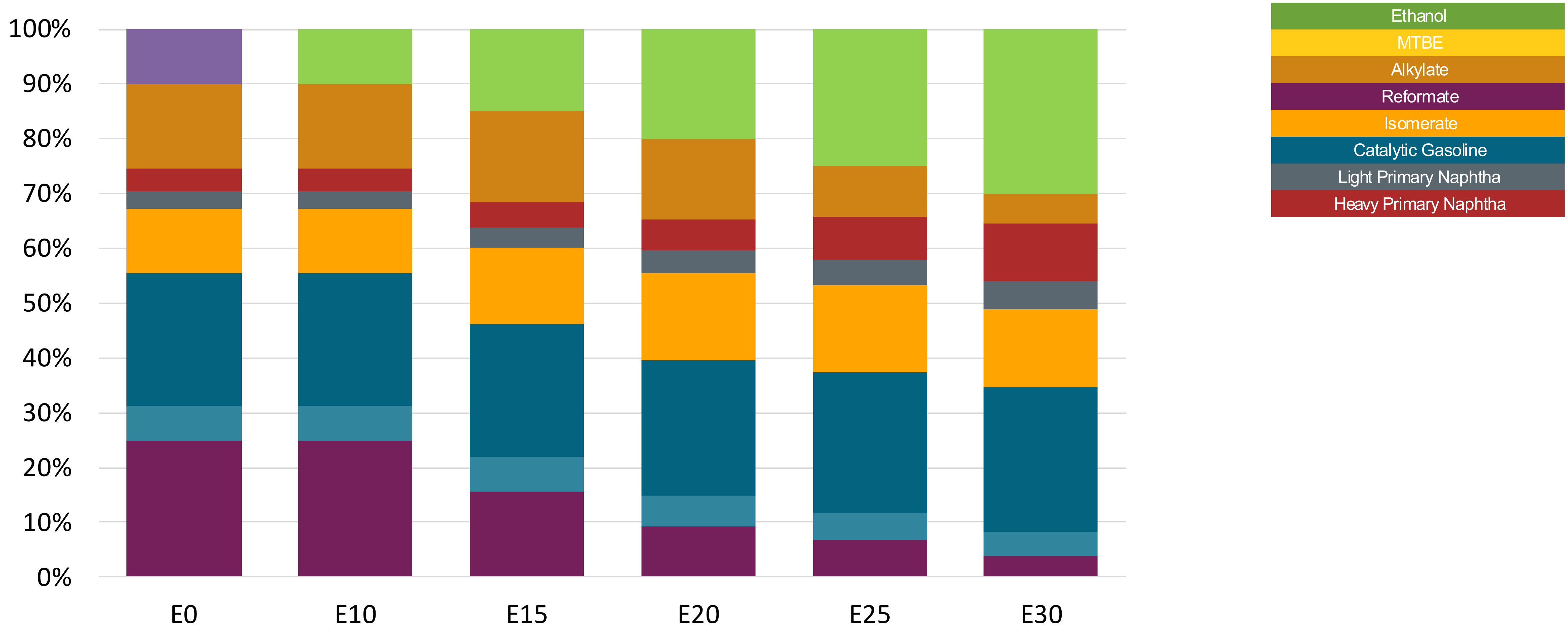


Available Blending Components



Fuente: ENAP, 2025

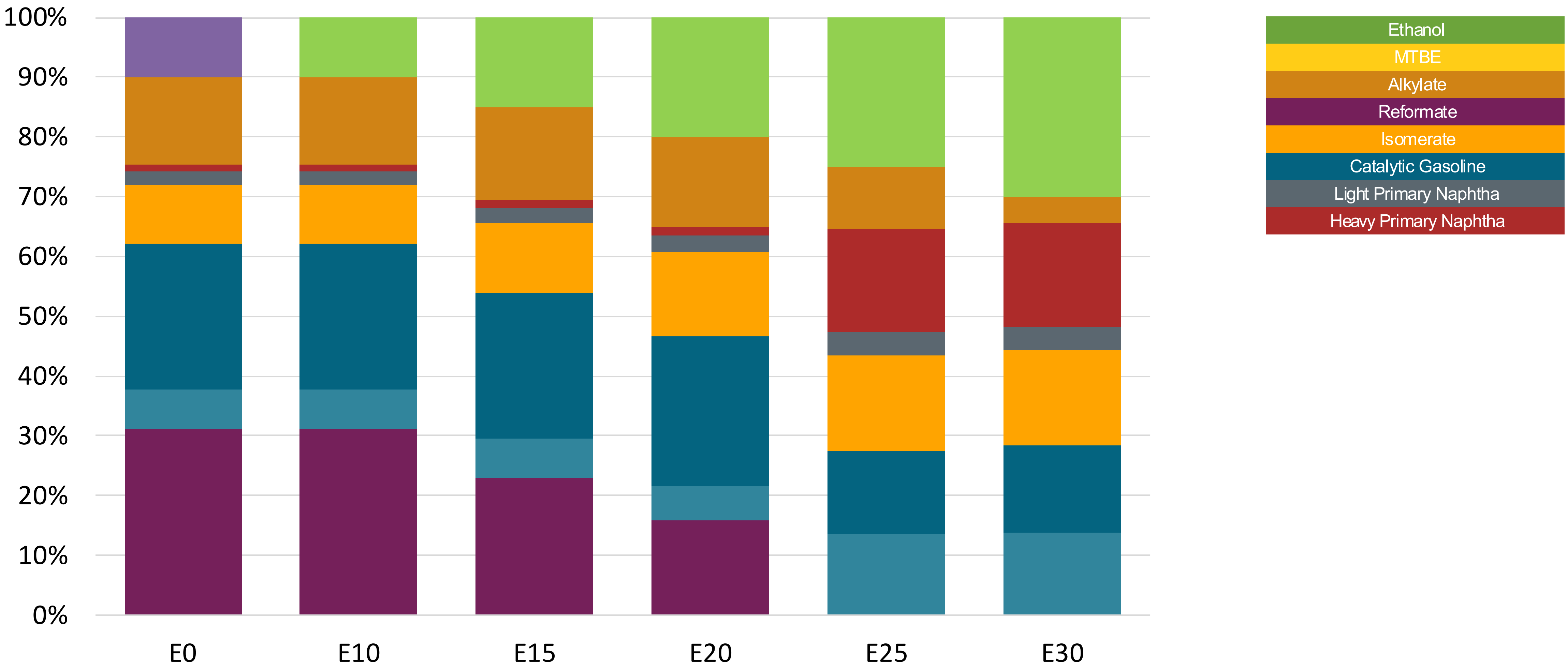
Chile – G93 RP Aconcagua – Constant Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	93	93	93	93	93	93
Price (US$/gal)	$2.601	$2.519	$2.430	$2.341	$2.249	$2.160

Source: Faro90, 2025

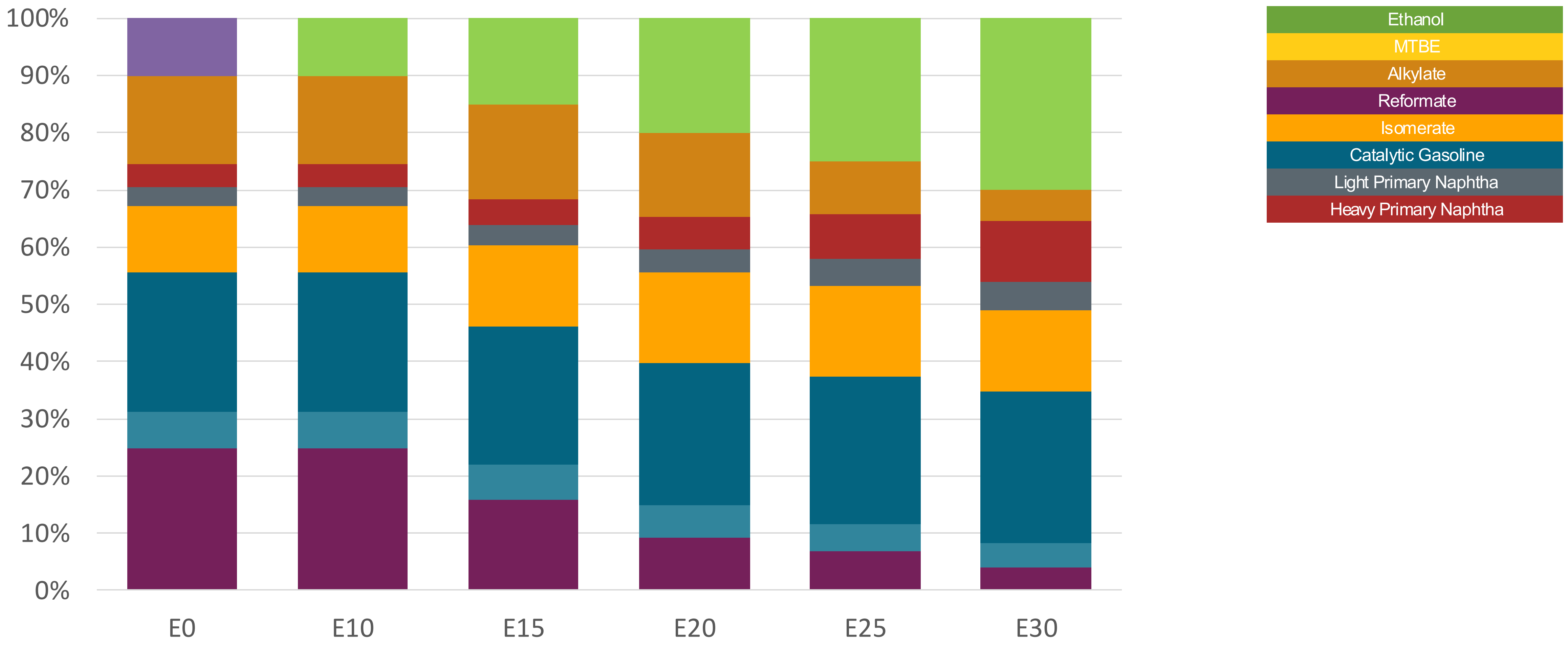
Chile – G97 RP Aconcagua – Constant Octane



Octane (RON)	97	97	97	97	97	97
Price (US$/gal)	$2.727	$2.645	$2.567	$2.487	$2.133	2.077

Source: Faro90, 2025

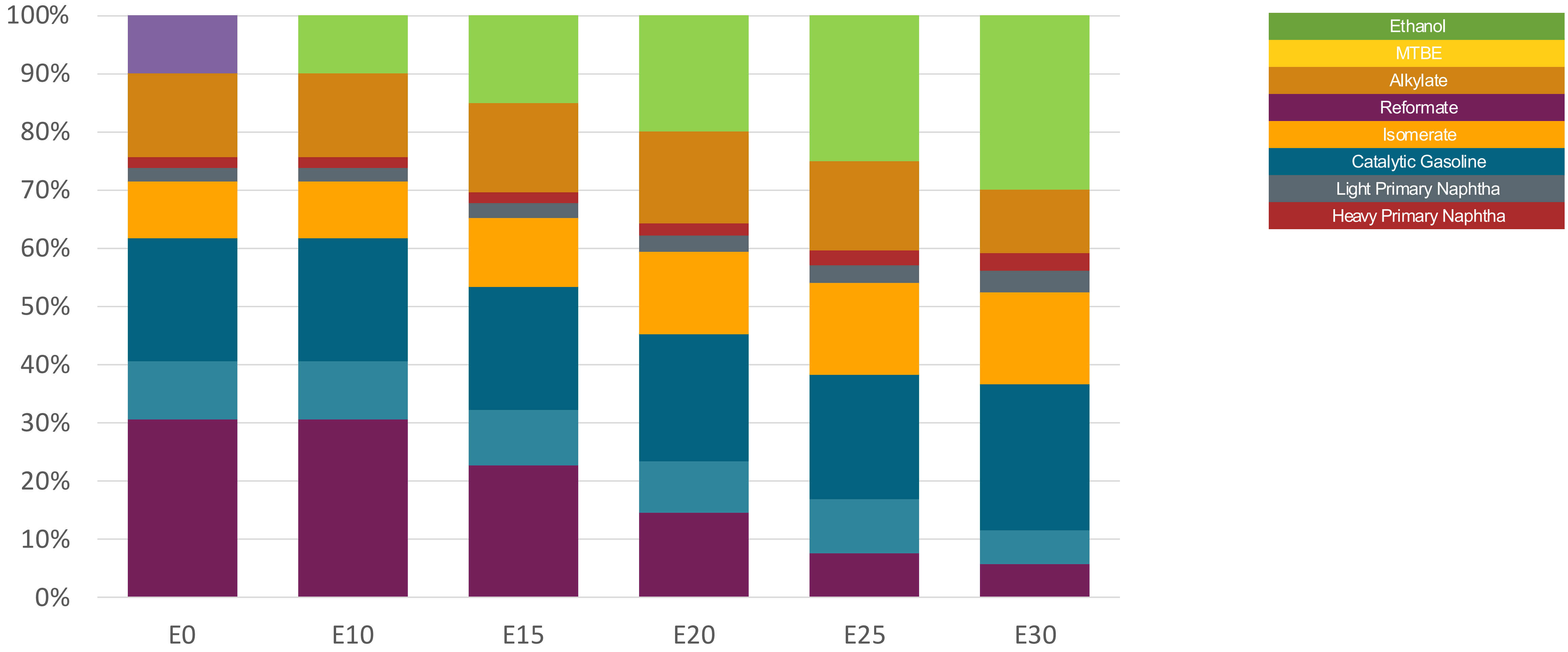
Chile – G93 RM Aconcagua – Constant Octane



	E0	E10	E15	E20	E25	E30
Octane (RON)	93	93	93	93	93	93
Price (US$/gal)	$2.601	$2.519	$2.430	$2.341	$2.249	$2.160

Source: Faro90, 2025

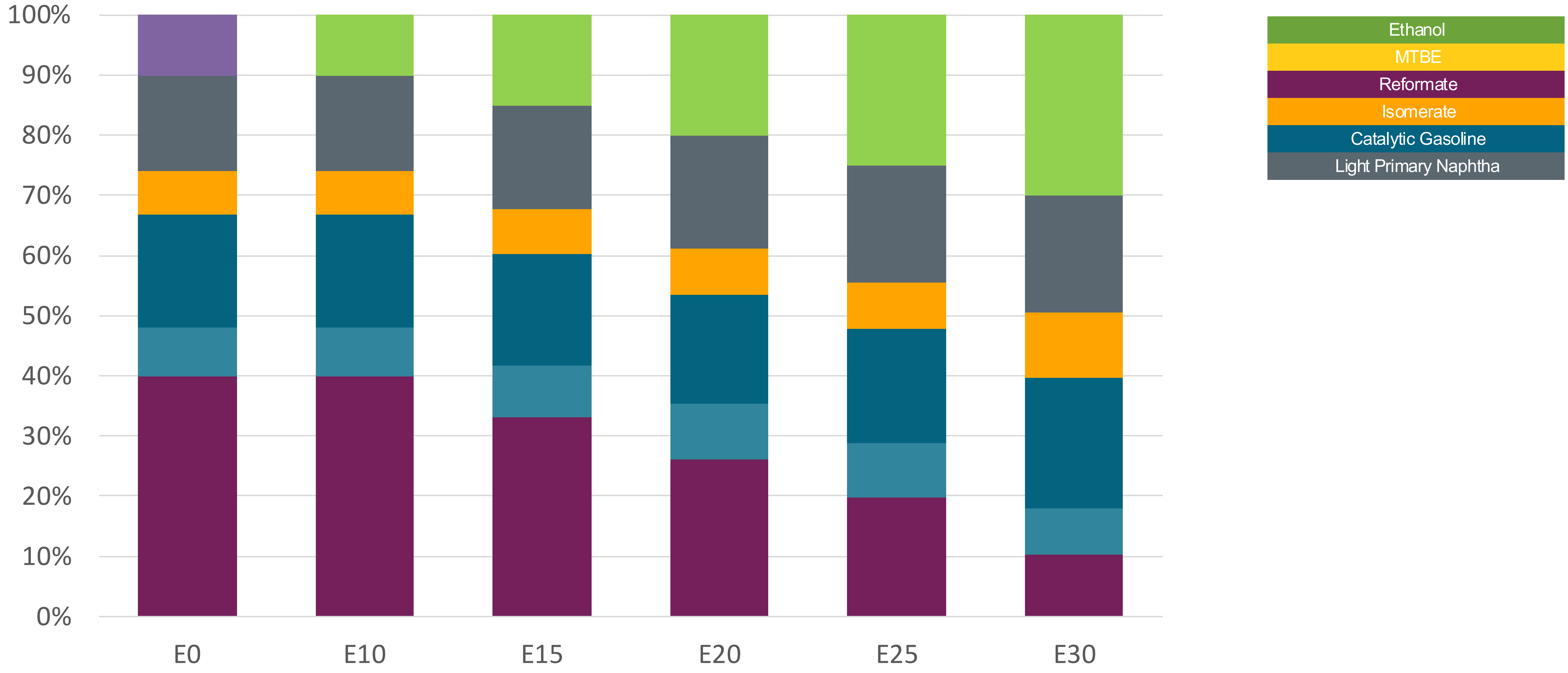
Chile – G97 RM Aconcagua – Constant Octane



Octane (RON)	97	97	97	97	97	97
Price (US$/gal)	$2.71	$2.63	$2.55	$2.47	$2.39	$2.32

Source: Faro90, 2025

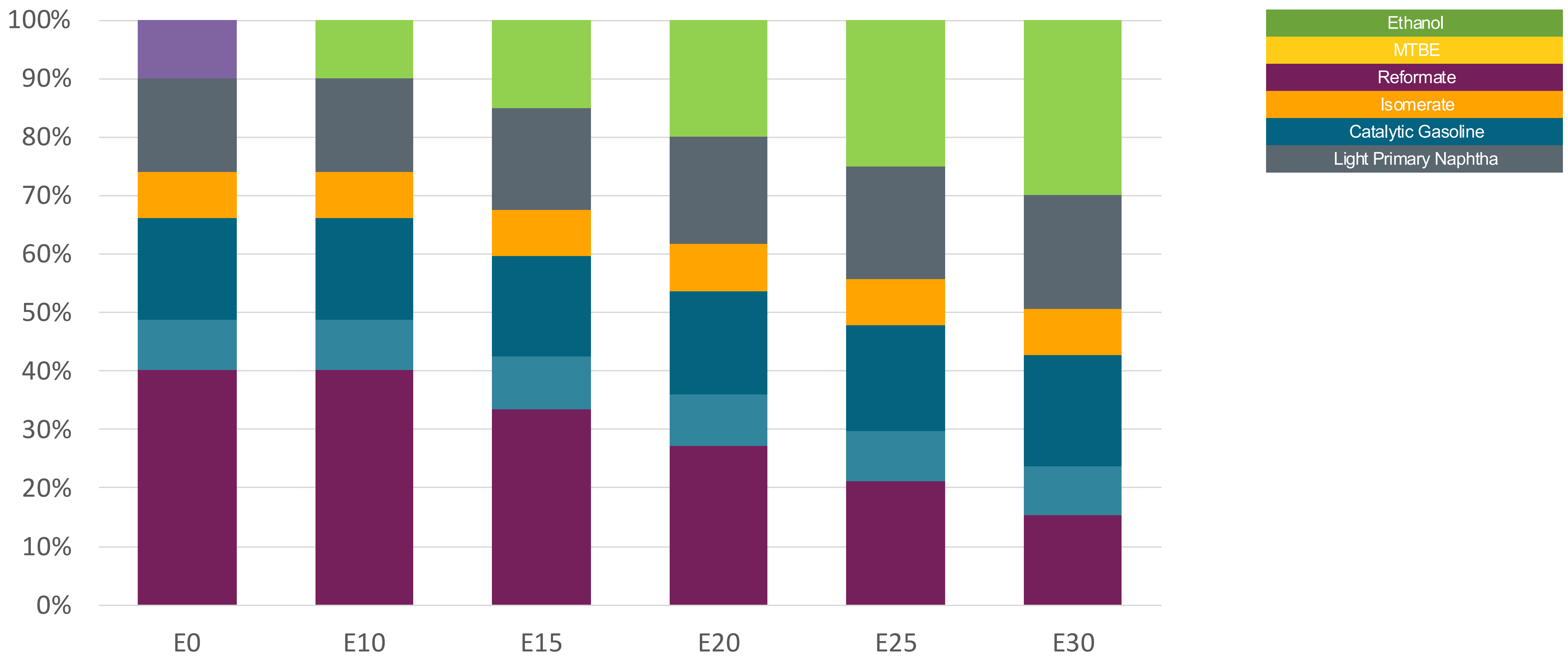
Chile – G93 RP Biobío – Constant Octane



Octane (RON)	93.0	93.0	93.0	93.0	93.0	93.0
Price (US$/gal)	$ 2.46	$ 2.38	$ 2.29	$ 2.21	$ 2.14	$ 2.06

Source: Faro90, 2025

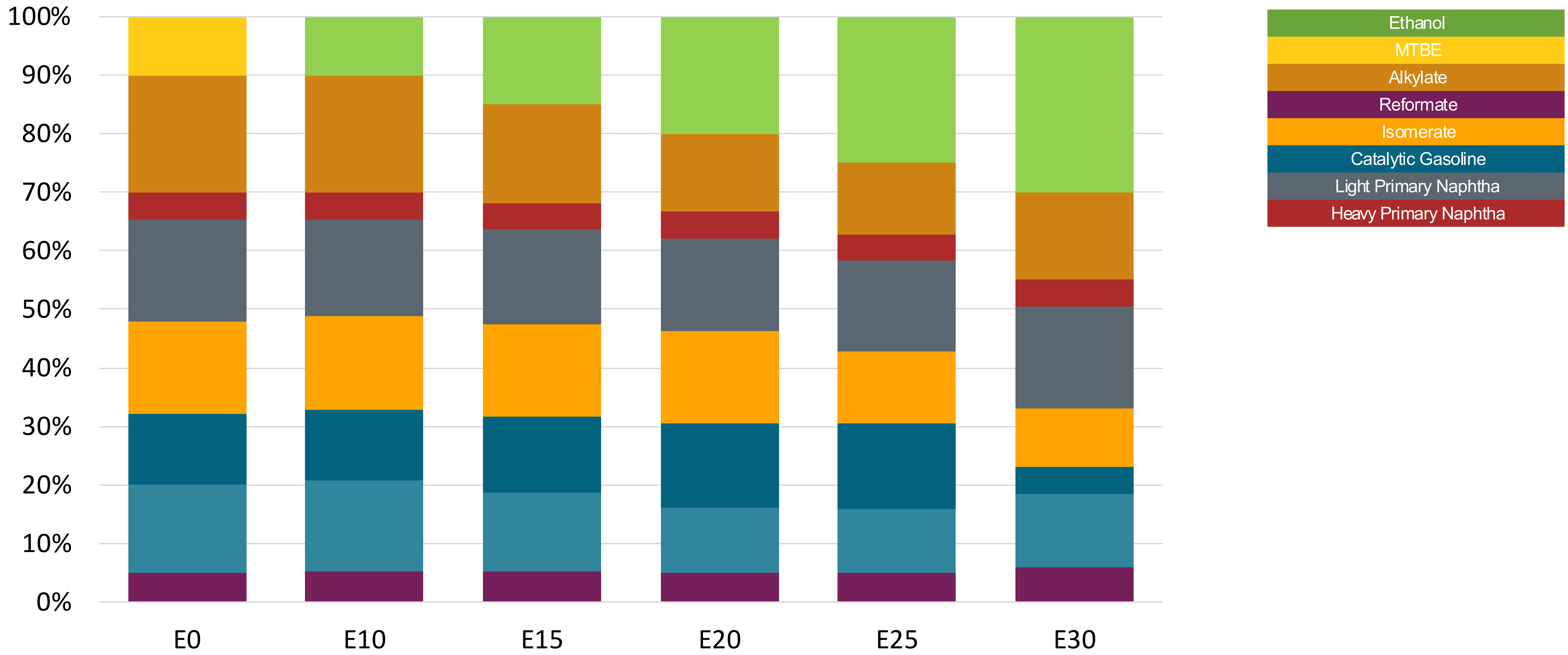
Chile – G97 RP Biobío – Constant Octane



Octane (RON)	97.0	97.0	97.0	97.0	97.0	97.0
Price (US$/gal)	$ 2.56	$ 2.47	$ 2.39	$ 2.31	$ 2.24	$ 2.17

Source: Faro90, 2025

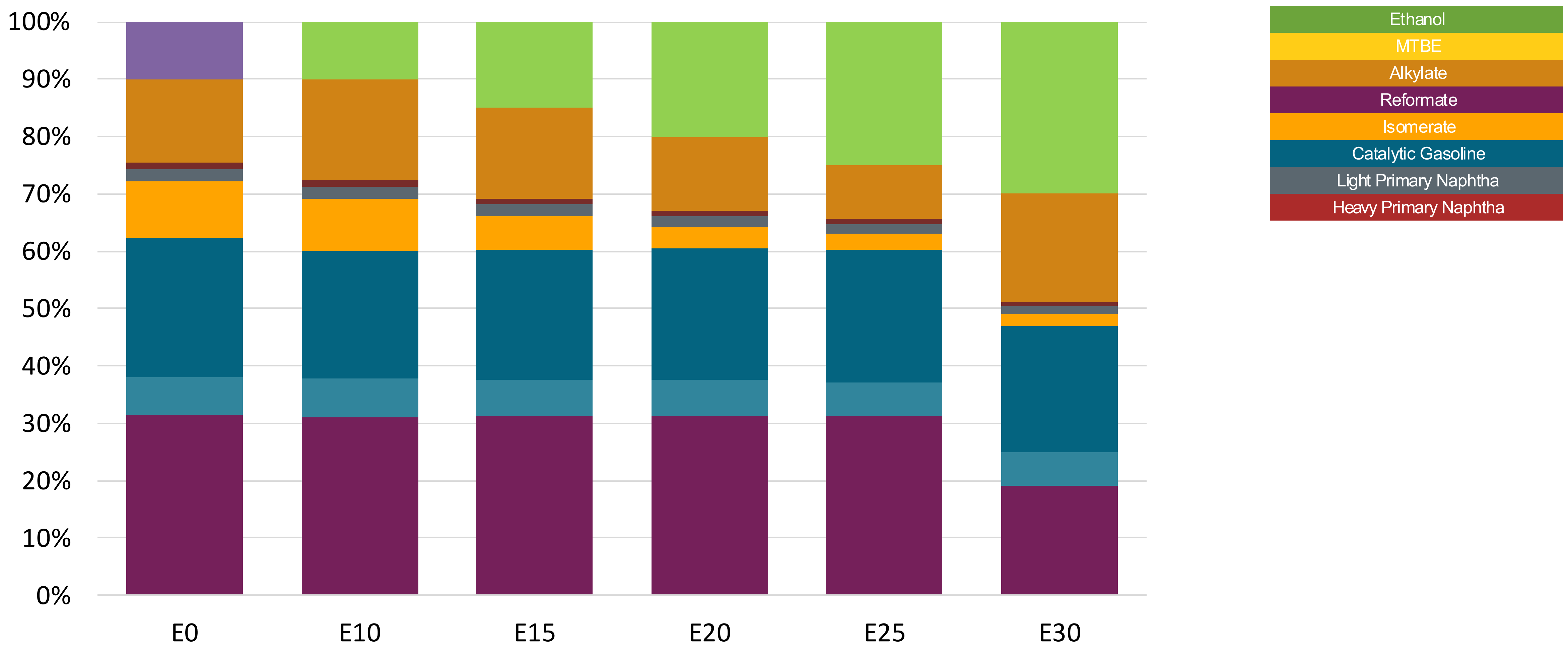
Chile – G93 RP Aconcagua – Octane Increment



Octane (RON)	93.0	95.1	97.5	100.1	102.0	104.3
Price (US$/gal)	$ 2.40	$ 2.40	$ 2.40	$ 2.40	$ 2.40	$ 2.40

Source: Faro90, 2025

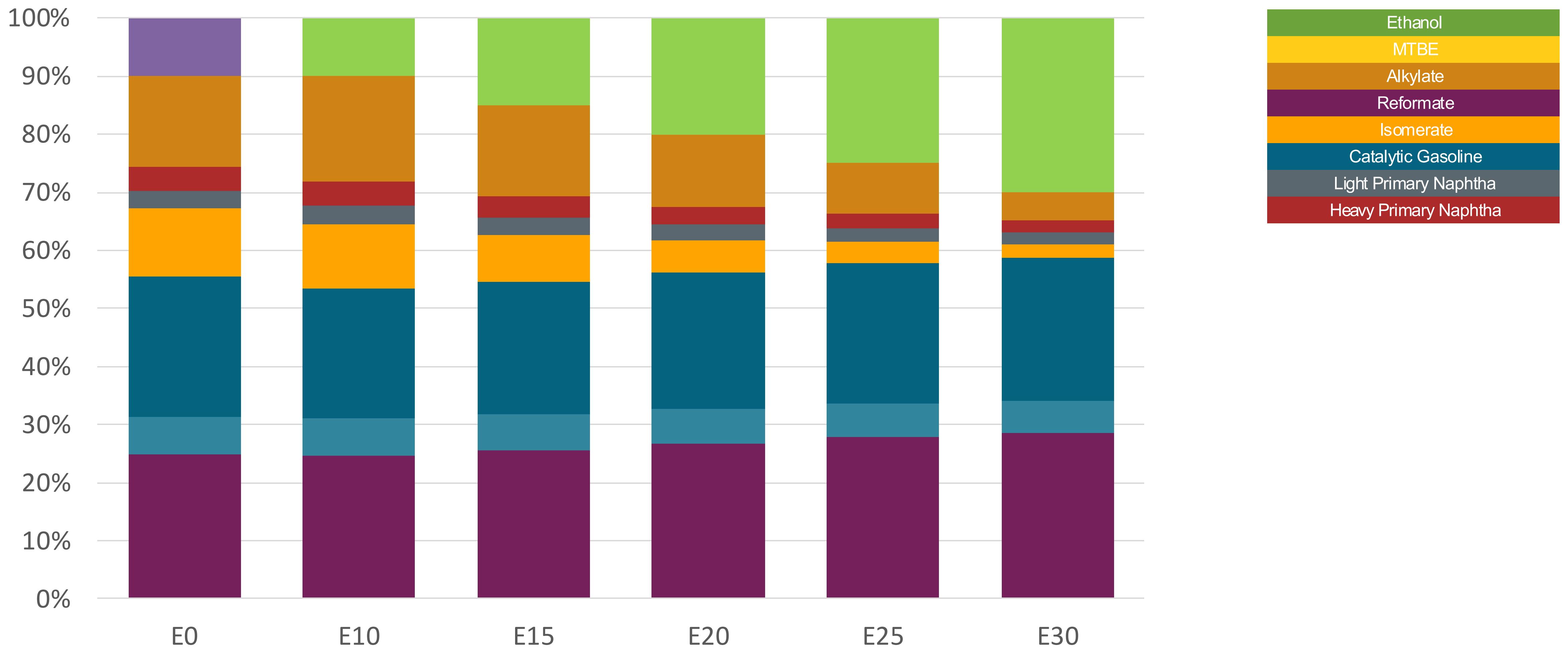
Chile – G97 RP Aconcagua – Octane Increment



Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.4
Price (US$/gal)	$2.73	$2.73	$2.73	$2.73	$2.73	$2.73

Source: Faro90, 2025

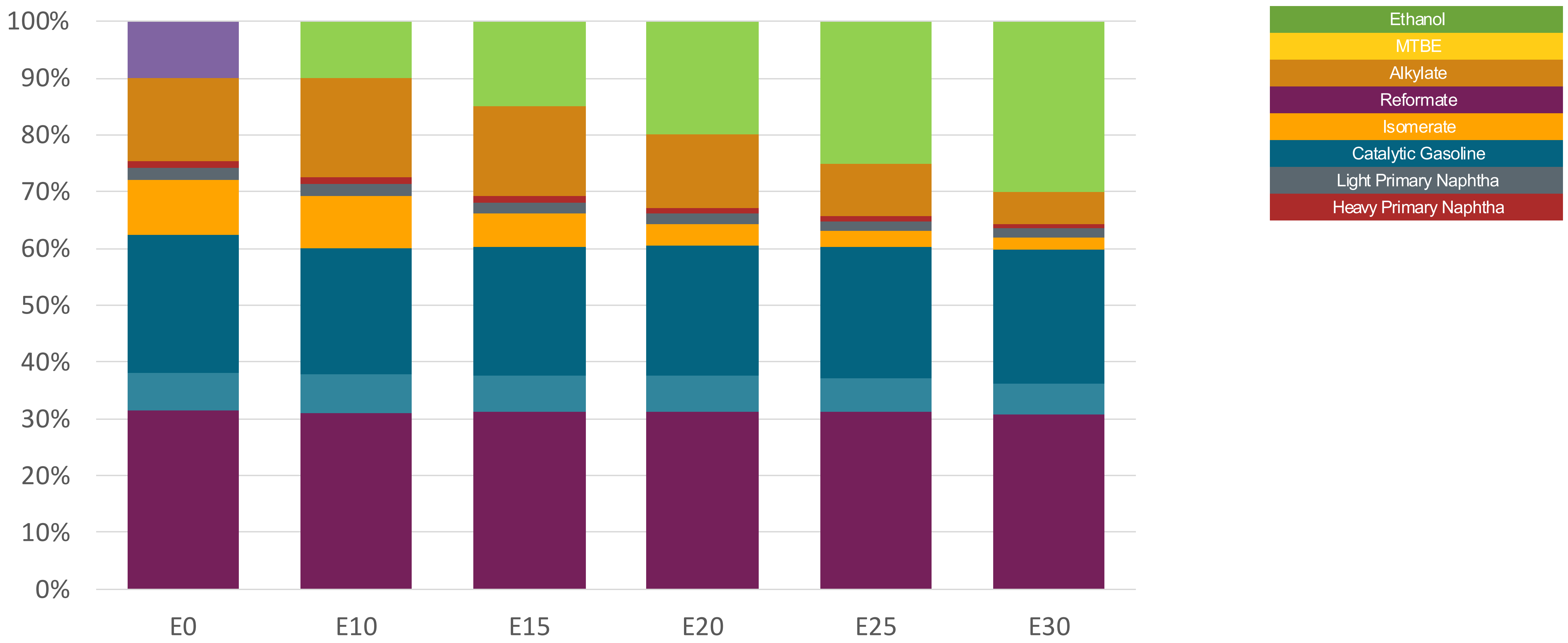
Chile – G93 RM Aconcagua – Octane Increment



Octane (RON)	93.0	96.2	98.1	100.0	102.1	104.3
Price (US$/gal)	$ 2.601	$ 2.601	$ 2.601	$ 2.601	$ 2.601	$ 2.601

Source: Faro90, 2025

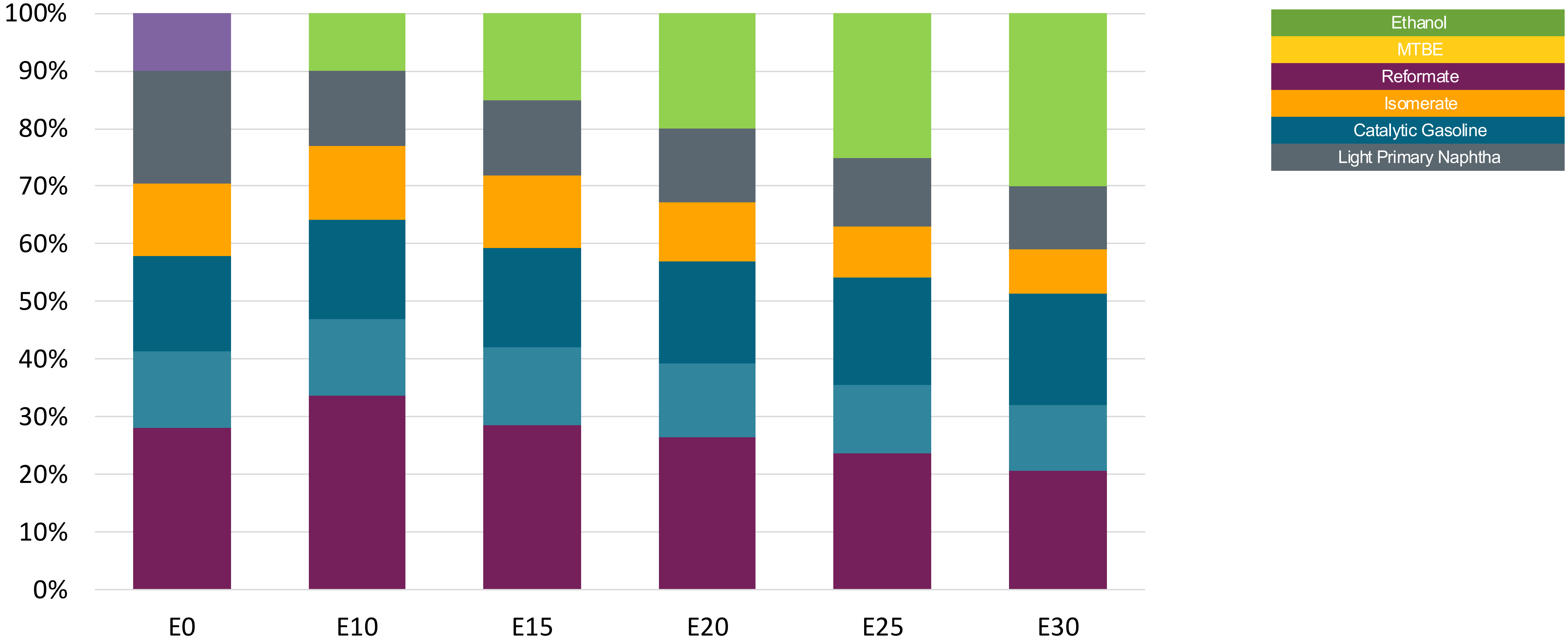
Chile – G97 RM Aconcagua – Octane Increment



	E0	E10	E15	E20	E25	E30
Octane (RON)	97.0	100.1	102.1	104.3	106.9	109.5
Price (US$/gal)	$ 2.73	$ 2.73	$ 2.73	$ 2.73	$ 2.73	$ 2.73

Source: Faro90, 2025

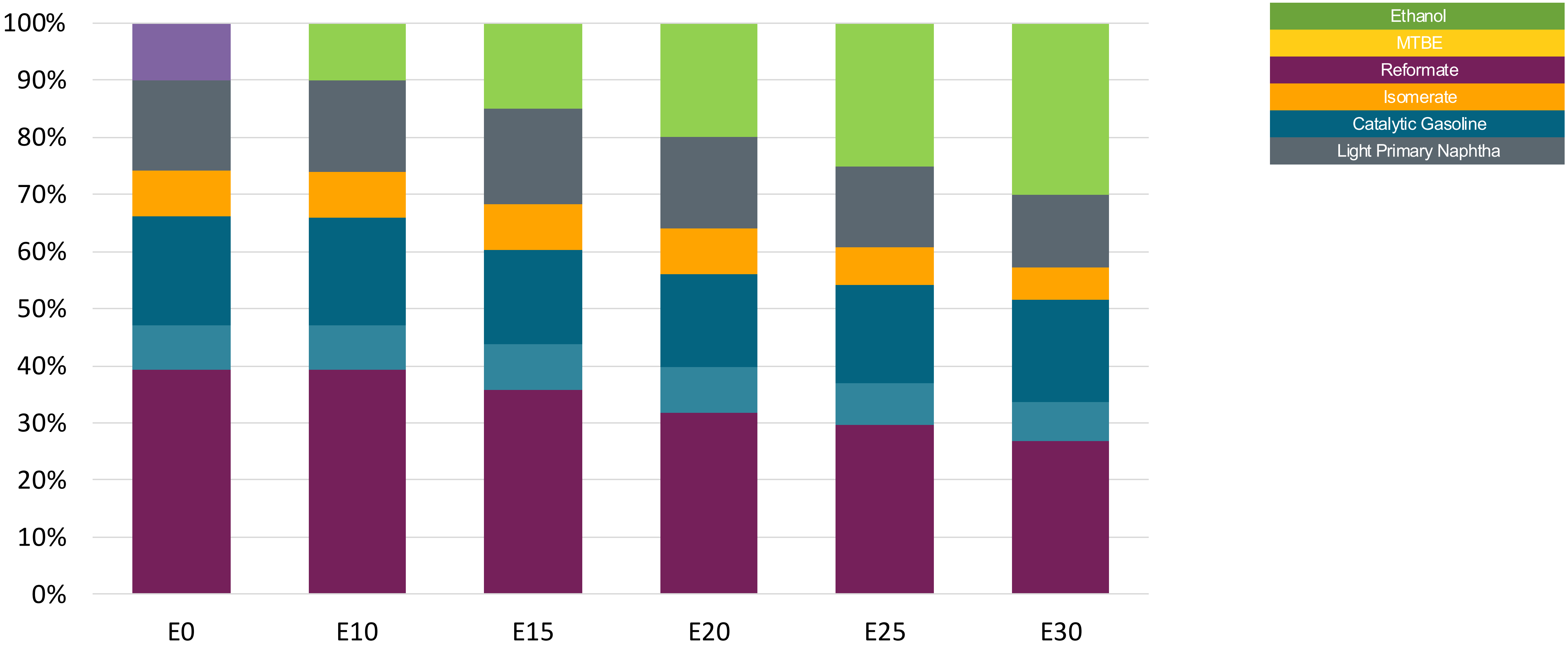
Chile – G93 RP Biobio – Octane Increment



Octane (RON)	93.0	93.0	95.9	98.2	100.4	102.7
Price (US$/gal)	$ 2.33	$ 2.33	$ 2.33	$ 2.33	$ 2.33	$ 2.33

Source: Faro90, 2025

Chile – G97 RP Biobío – Octane Increment



Octane (RON)	97.0	101.0	103.7	107.1	108.9	111.1
Price (US$/gal)	$ 2.558	$ 2.558	$ 2.558	$ 2.558	$ 2.558	$ 2.558

Source: Faro90, 2025

Vehicle Emission Impact for Ethanol Gasoline Blending

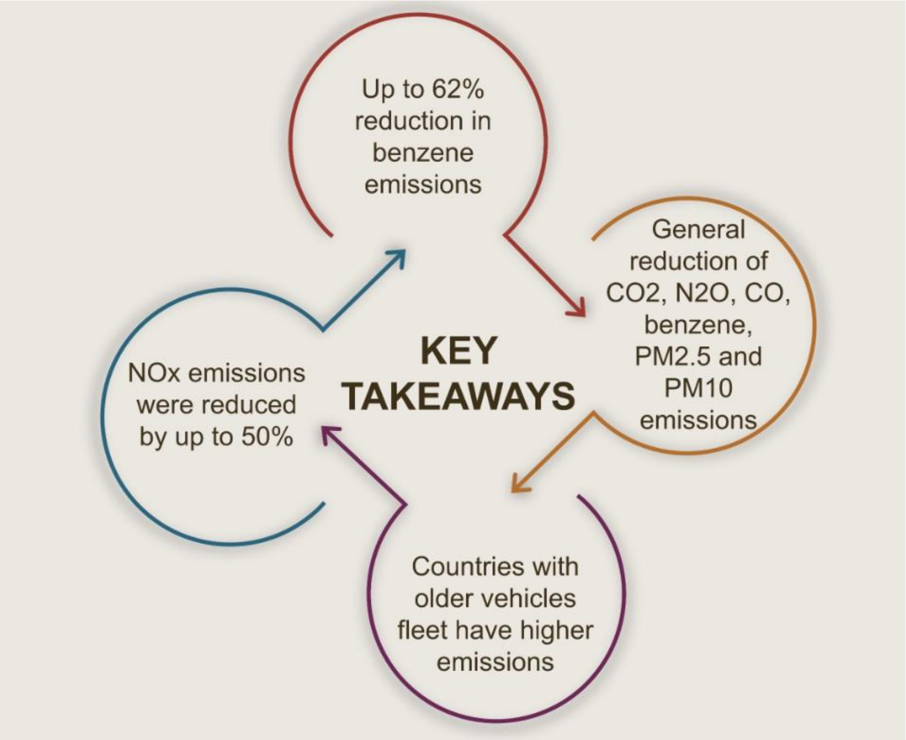
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Chile – Vehicular Fleet

Tipo	Peso	Motor	Escape	Recuperator	Age	Index	Number of Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	+8años	29	42,608
		Single FI	3-Ways		+8años	83	298,823
		Multi FI	3-Ways		+8años	119	1,120,144
			Euro III		+8años	191	690,653
			Euro IV		+8años	200	787,336
			Euro V		+8años	1253	464,935
					4-8años	1252	803,943
			Euro VI		-4años	1251	415,697
		Hybrid	Hybrid		4-8años	208	898
					-4años	207	3,650
		Electric					1,340
	Medium	Carburator	3-Ways	PCV Tank	+8años	32	246,934
		Single FI	3-Ways		+8años	86	337,941
		Multi FI	3-Ways		+8años	122	393,950
			Euro III		+8años	194	167,702
			Euro IV		+8años	203	291,291
			Euro V		+8años	1256	89,733
					4-8años	1255	263,841
			Euro VI		-4años	1254	236,646
		Hybrid	Hybrid		4-8años	211	0
					-4años	210	495
		Electric					415
Sml Engines	Ligeros	4-Cycle	Euro II	PCV	+8años	1235	33,031
					+8 años	1234	999
			Euro III		+8 años	1243	145,315
					-4 años	1242	
			Euro IV		4-8años		21,990
			Euro V		4-8años		85,681
		Multi FI	Hibrido				0
		Electric	-		-8años		258
Truck	Pesados	Carburator	3-Ways	PCV	+8años	863	105,504
		Fuel-Injection	3-Ways		+8años	908	177,934
			Euro III		+8años	944	35,057
			Euro IV		+8años	953	45,369
			Euro V		+8años	962	25,786
					4-8años	961	12,630
			Euro VI		-4años	960	31,201

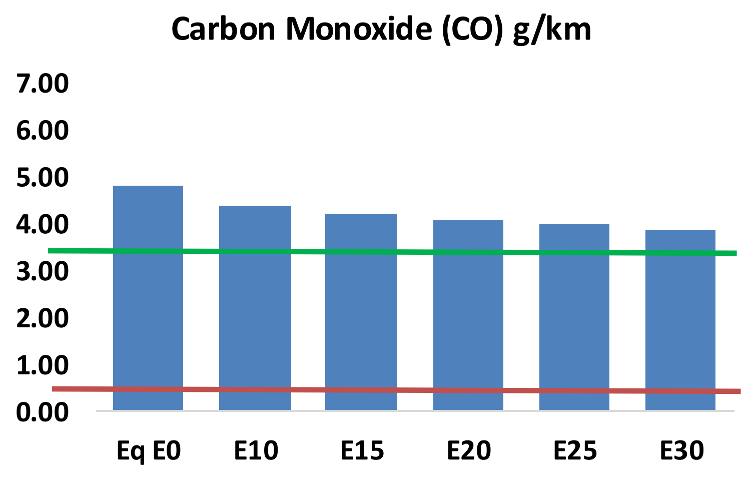
Vehicular Fleet: **7,379,730 vehicles that use gasoline**
Average Age: **10 year**

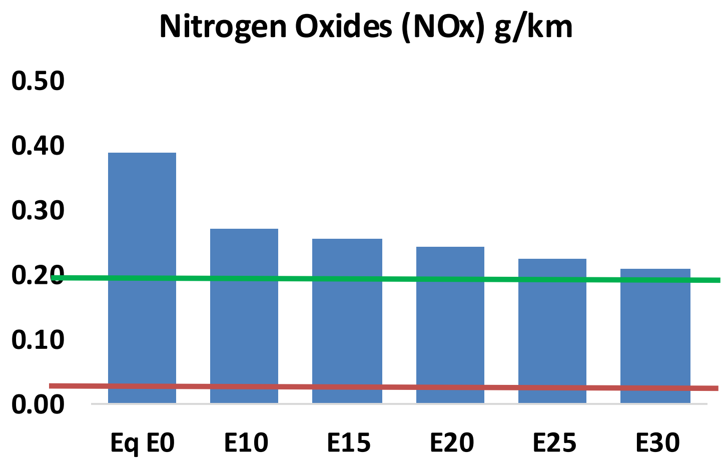
Main Technology: **Motor Multi-Fuel-Injection, Euro V, PCV Tank**

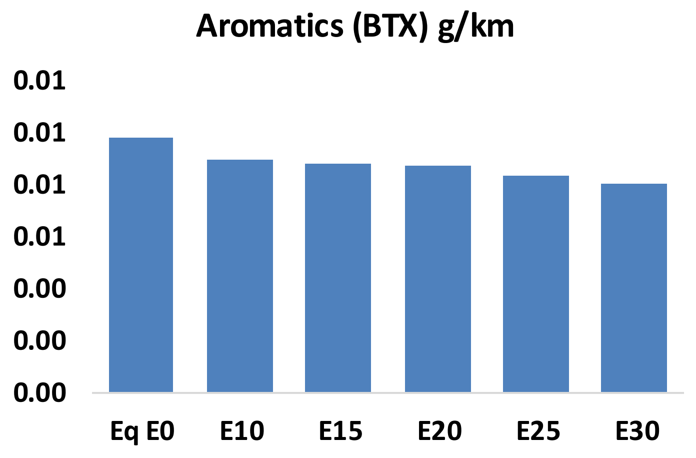
Source: INE, 2024

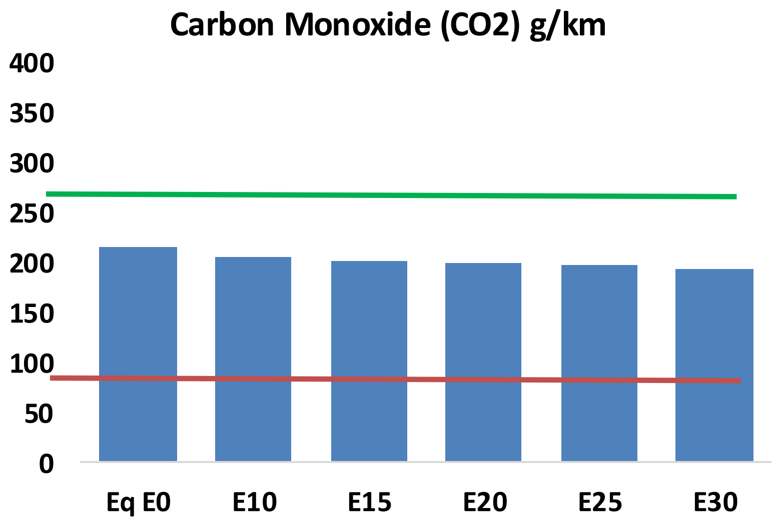
Chile – Gasoline Vehicular Fleet Emissions

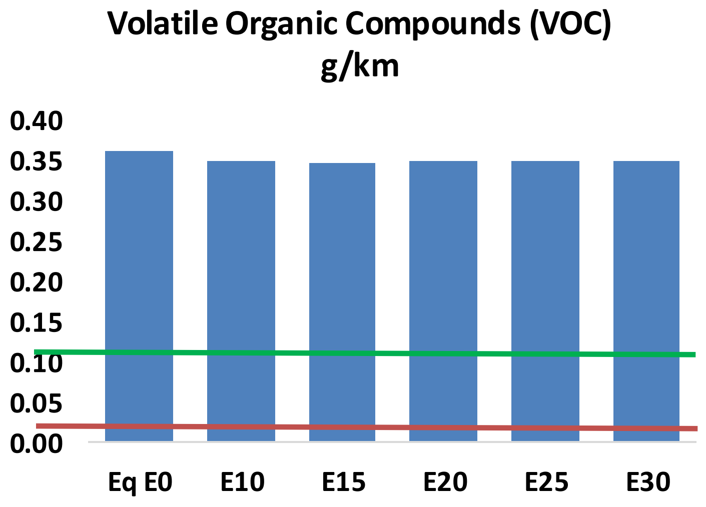
United States
Euro 6

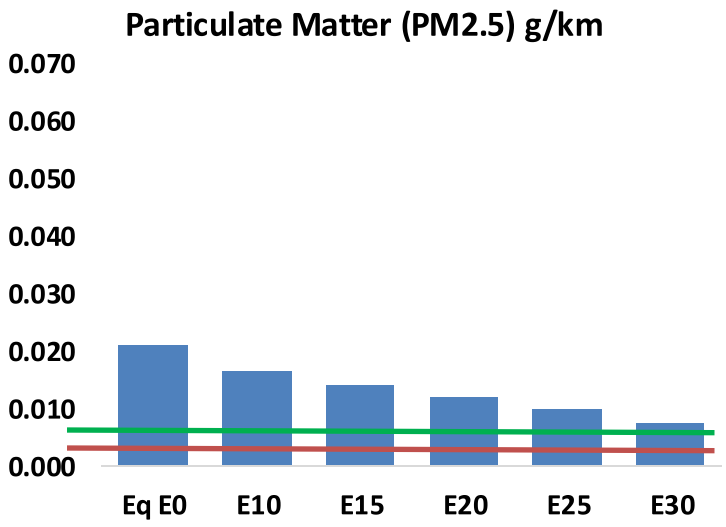












Source: Faro 90

Chile – Gasoline Vehicular Fleet Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	490,224	468,404	446,583	431,799	418,345	408,431	394,171	-9%	-15%
CO2	22,058,884	21,491,656	20,955,940	20,534,616	20,326,128	20,205,060	19,832,623	-5%	-8%
VOC	36,934	36,273	35,612	35,496	35,574	35,694	35,645	-4%	-4%
NOx	39,757	29,818	27,830	26,230	24,737	23,091	21,350	-30%	-38%
SOx	268	247	227	210	194	176	157	-15%	-28%
NH3	7,206	7,135	7,064	7,097	7,177	7,233	7,280	-2%	0%
N2O	392	389	388	390	398	402	407	-1%	2%
CH4	8,281	8,281	8,281	8,405	8,587	8,745	8,894	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	448	441	434	433	435	438	438	-3%	-3%
Acetaldehyde	801	919	1,349	2,055	2,799	3,199	3,779	68%	249%
Formaldehyde	2,301	2,237	2,608	3,062	3,208	3,549	3,863	13%	39%
Benzene	1,000	961	911	896	889	850	823	-9%	-11%
PM2.5	1,341	1,191	1,041	904	767	623	472	-22%	-43%
PM10	811	721	630	547	464	377	286	-22%	-43%

Source: Faro 90

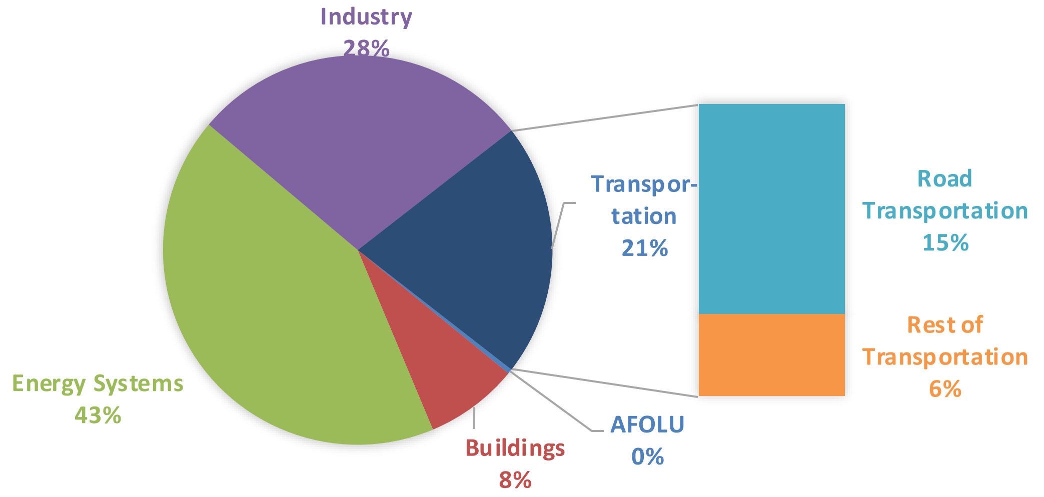
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

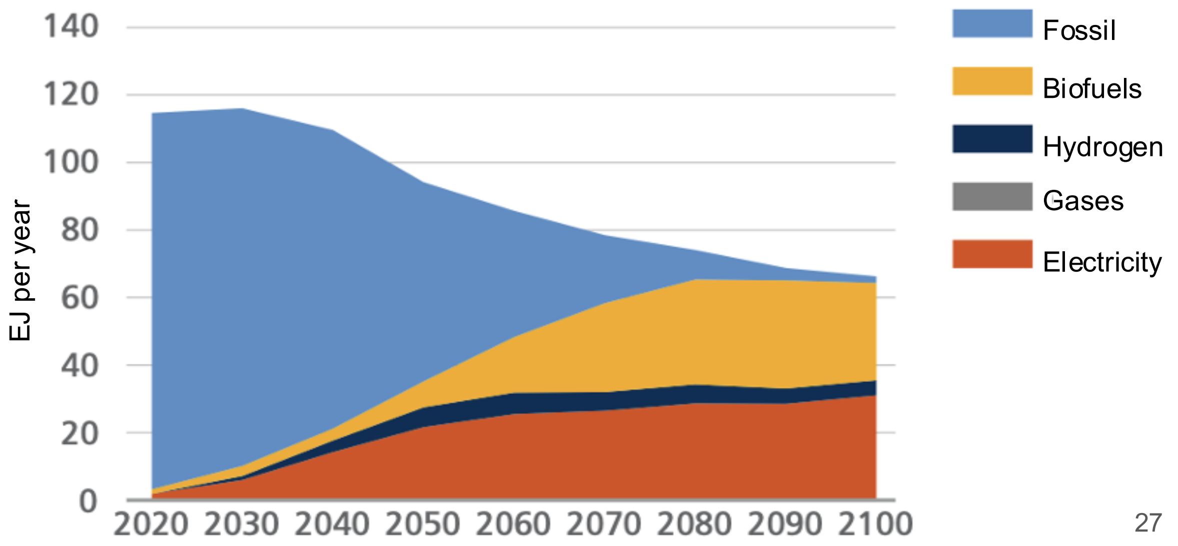
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

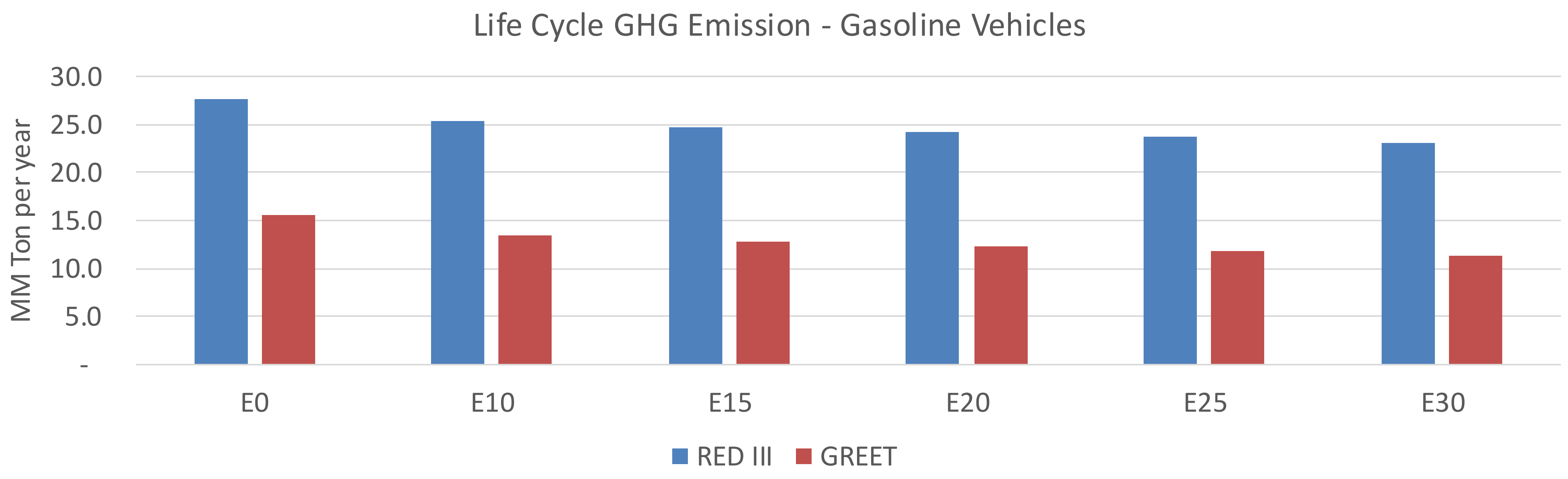
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Chile – Gasoline Vehicle Fleet GEI Emissions



Country Emissions 2024 (MMT/year)	118.1
2035 Target (MMT/year)	90.1
Delta	28.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	13.8%	17.3%	20.4%	23.4%	27.1%
RED II/III	0.0%	8.2%	10.4%	12.3%	14.2%	16.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.6%	9.6%	11.3%	13.0%	15.0%
RED II/III	0.0%	8.1%	10.3%	12.1%	14.0%	16.1%

Source: Greet, RED II/III, climateactiontracker.org, F90